Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

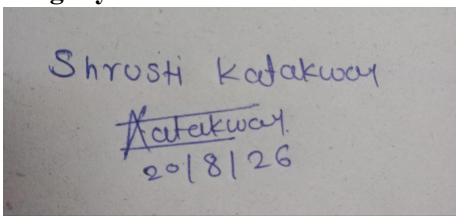
Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.



Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**HCPA**
Semester	**5th**	**Course Code**	**23ADS1524**
Roll No	**119**	**Name of Student**	**Shrusti Katakwar**

Practical Number	**02**
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	**Write a basic MPI program for point-to-point communication.**
Problem Definition	**Write a basic MPI program for point-to-point communication.**
Theory (100 words)	MPI (Message Passing Interface) is a standard used for communication between multiple processes in a parallel computing environment. In MPI, each process has a unique rank, which identifies it within a communicator. Point-to-point communication means that one process directly sends a message to another process. The two basic MPI functions used are: • `MPI_Send()` – used by the sender process to send data. • `MPI_Recv()` – used by the receiver process to receive data. For example, process 0 can send a message to process 1. The receiver specifies the rank of the sender from which it expects the message. MPI provides efficient communication between processes running on the same or different systems.

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Procedure and Execution

(100 Words)

Step for Implementation:

Include the MPI header file using `#include <mpi.h>`.
Initialize the MPI environment using `MPI_Init()`.
Obtain the rank of each process using `MPI_Comm_rank()`.
Obtain the total number of processes using `MPI_Comm_size()`.
Check that at least two processes are available.
Make **Process 0** the sender.
Store the value to be transmitted in a variable.
Use `MPI_Send()` in Process 0 to send the data to Process 1.
Make **Process 1** the receiver.
Use `MPI_Recv()` in Process 1 to receive the data from Process 0.
Display the sent and received values.
Terminate the MPI environment using `MPI_Finalize()`.
Compile and execute the program using multiple MPI processes.

Code:

```
#include <mpi.h>
#include <stdio.h>

int main(int argc, char* argv[]) {
    int rank, size;
    int number;

    // Initialize MPI environment
    MPI_Init(&argc, &argv);

    // Get the rank and size
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    if (size < 2) {
        fprintf(stderr, "Need at least two processes.\n");
        MPI_Abort(MPI_COMM_WORLD, 1);
    }

    if (rank == 0) {
        number = 42; // Message to send
        MPI_Send(&number, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
        printf("Process 0 sent number %d to process 1\n", number);
    } else if (rank == 1) {
        MPI_Recv(&number, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        printf("Process 1 received number %d from process 0\n", number);
    }

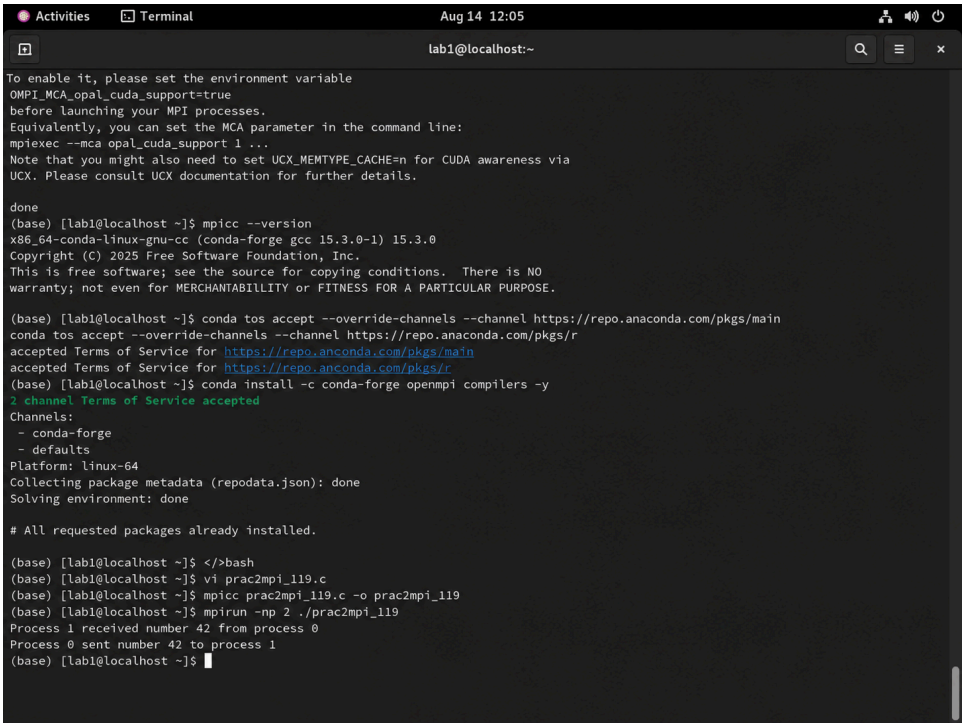
    // Finalize MPI
    MPI_Finalize();
    return 0;
}
~
~
~
-- INSERT --
```

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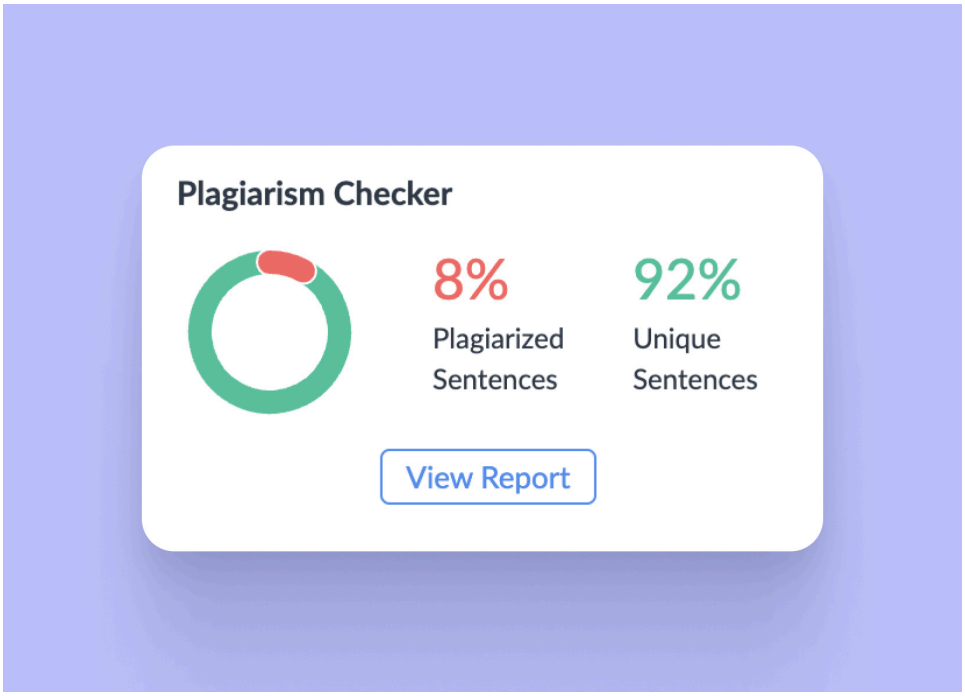
	Output: 
Output Analysis	Process 0 acts as the sender and sends the value `100`. Process 1 acts as the receiver and receives the same value. `MPI_Send()` performs the point-to-point sending operation. `MPI_Recv()` performs the corresponding receiving operation. The matching source rank and message tag ensure that Process 1 receives the intended message. The output confirms that data was successfully transferred from Process 0 to Process 1. Thus, the program demonstrates basic point-to-point communication using MPI.
Link of student Github profile where lab assignment has been uploaded	
Conclusion	The MPI program successfully demonstrates point-to-point communication between two processes. Process 0 sends data to

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	Process 1 using `MPI_Send()`, and Process 1 receives the data using `MPI_Recv()`. Thus, MPI provides an efficient mechanism for direct communication between processes in a parallel computing environment.
Plag Report (Similarity index < 12%)	
Date	**20/08/26**